UPS REPRICE PLATFORM · TECHNICAL DOCUMENTATION

UPS Address Validation API Setup Guide

Replacing the built-in residential / commercial decision with the UPS address classification service

Purpose	Deploying the proxy, configuring credentials, verifying it, the interface contract, client setup, and known limits
Audience	Administrators (sections 1–4, 6–7); everyday users (section 5)
Prerequisites	A UPS developer account, access to a Supabase project, the Supabase CLI (section 2)
Project ref	Your own Supabase project Reference ID (how to find it: 2.2). Written throughout as `<your-project-ref>`
Proxy source	`supabase/functions/ups-address/index.ts`

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1 Overview

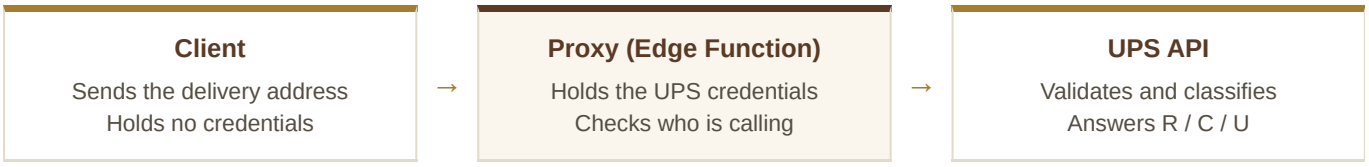
1.1 What it does

By default the system decides whether each shipment is residential or commercial from the billing codes, the service name and the invoice history. With this feature on, the delivery address is sent to the official UPS address validation service instead, which answers residential or commercial.

Residential and commercial rates differ, so that answer changes the money. Read section 7 before turning it on.

1.2 Architecture

The browser cannot call UPS directly: UPS requires OAuth and does not allow cross-origin access (CORS). A proxy is therefore deployed as a Supabase Edge Function, holds the credentials, and makes the call on the browser's behalf.



The answers are `R` residential, `C` commercial, `U` undetermined. On `U`, or on any failure, pricing falls back to the built-in logic and carries on — nothing stalls.

1.3 Where credentials live

The UPS Client ID and Client Secret exist only as environment variables on the proxy. The client settings screen has no field for them, deliberately.

> **Why there is no field**
>
> Every field on the settings screen is written into `CFG.raw`, and that object goes into browser localStorage, into the account's synced settings row, and into any exported settings file. A credential placed there would spread through all three. On top of that, CORS stops the browser from reaching UPS at all, so a credential entered there could not be used even if it were safe to store.

2 Prerequisites

Three things are needed before section 3. They come from different places and serve different purposes; without any one of them, deployment cannot finish.

2.1 UPS developer account

What it is for: obtaining the Client ID and Client Secret that authorise calls to UPS, and subscribing to the Address Validation product. UPS issues credentials per application, and one set of credentials draws on one UPS account's quota.

Item	Detail
Where	`developer.ups.com` → Apps → Add App
Requires	A working UPS account number — one that actually ships and is invoiced
Product to add	**Address Validation**. Without it the credentials are still valid, but calls come back `502 UPS 403`
What you get	A Client ID and a Client Secret. The Secret is shown once, at creation — save it then
Environments	Test `wwwcie.ups.com`, production `onlinetools.ups.com`. Both use the same credentials

> **One UPS account, one set of credentials**
> Usage is billed to the UPS account the credentials belong to. Several UPS accounts each need their own application and their own credentials; sharing one set makes quota and billing impossible to attribute.

2.2 Supabase project access

What it is for: the proxy runs as an Edge Function on a Supabase project, and that project holds the UPS credentials.

Item	Detail
Role needed	Owner or Developer on the project. Read access cannot deploy functions or write environment variables
Project ref	The `Reference ID` under `Settings → General` — the subdomain in your endpoint URL. Commands in this guide write it as `<your-project-ref>`; substitute your own
Key you need	The anon (publishable) key, for the test in 4.2. This key is public by design and ships in your `auth-config.js`; what protects the data is row-level security and function grants, not the key's secrecy. Never substitute the `service_role` key
How to check	If `Edge Functions` appears in the console sidebar, your role is sufficient

2.3 Supabase CLI

What it is for: deploying the function and writing environment variables. Neither can be done from the web console; the CLI is required.

Platform	Install
macOS	`brew install supabase/tap/supabase`
Windows	`scoop install supabase`
Any (Node)	`npm install -g supabase`

Check the version and sign in:

```
supabase --version
supabase login
```

`supabase login` opens a browser to authorise; afterwards the CLI acts with that account's permissions. Once per computer.

2.4 Readiness checklist

All four must hold before section 3.

#	Item	How to confirm
1	The UPS application exists and subscribes to Address Validation	The product is listed on the app's detail page and shows as approved
2	The Client ID and Client Secret are saved	You can read both back. A lost Secret must be regenerated
3	You can deploy to the Supabase project	Edge Functions is visible in the console
4	The CLI is installed and signed in	`supabase --version` prints something and `supabase login` has completed

3 Deployment

3.1 Obtain UPS API credentials

1. Create an application at `developer.ups.com`.
2. Add **Address Validation** to that application's products.
3. Take the Client ID and Client Secret.

> **The product subscription is not optional**
>
> The proxy calls `POST /api/addressvalidation/v2/3`. The trailing `3` is request option 3 — validation *and* classification. An application subscribed only to Rating or Tracking does not cover it, and verification will return `502 UPS 403`.

3.2 Link your Supabase project

```
# authorise the CLI (opens a browser)
supabase login

# from the repository root; writes supabase/config.toml
supabase link --project-ref <your-project-ref>
```

The repository ships without `config.toml`, so linking comes before the first deploy.

3.3 Set the environment variables

```
supabase secrets set \
  UPS_CLIENT_ID=<client-id> \
  UPS_CLIENT_SECRET=<client-secret> \
  UPS_BASE=https://wwwcie.ups.com
```

Variable	Detail
`UPS_CLIENT_ID`	Required
`UPS_CLIENT_SECRET`	Required
`UPS_BASE`	Optional. Defaults to `https://onlinetools.ups.com` (production). Point the first deployment at `https://wwwcie.ups.com` (the CIE test environment) so it spends no production quota
`SUPABASE_URL` `SUPABASE_ANON_KEY`	Do not set these — the platform injects them

3.4 Deploy the function

```
supabase functions deploy ups-address
```

Changing an environment variable takes effect immediately; no redeploy needed.

4 Verification

The proxy answers different failures with different status codes, so working through them in order tells you which layer is wrong. Run 4.1 to 4.3 in sequence; do not skip ahead.

4.1 Confirm the deployment

Call it with no credentials at all:

```
curl -i -X POST \
  "https://<your-project-ref>.supabase.co/functions/v1/ups-address" \
  -H "Content-Type: application/json" -d "{}"
```

Response	Reading
`401 not signed in`	Deployed. The proxy is correctly refusing an unauthenticated request
`404`	Not deployed — back to 3.4

4.2 Get an access token

In the Supabase console, `Authentication → Users → Add user`, create a test account (email and password, marked confirmed). Then:

```
curl -X POST \
  "https://<your-project-ref>.supabase.co/auth/v1/token?grant_type=password" \
  -H "apikey: <anon key, see auth-config.js>" \
  -H "Content-Type: application/json" \
  -d '{"email":"<email>","password":"<password>"}'
```

Take `access_token` from the response.

4.3 Confirm the endpoint

```
curl -X POST \
  "https://<your-project-ref>.supabase.co/functions/v1/ups-address" \
  -H "Authorization: Bearer <access_token>" \
  -H "Content-Type: application/json" \
  -d '{"address":"1600 Pennsylvania Ave NW","city":"Washington",
       "state":"DC","postal":"20500","country":"US"}'
```

Response	Reading and what to do
`200 {"class":"R"|"C"|"U"}`	Working. Continue to 4.4
`500 UPS credentials not configured`	Environment variables not set — back to 3.3
`502 UPS 401`	Wrong UPS credentials — check the Client ID and Secret
`502 UPS 403`	The application does not subscribe to Address Validation — back to 3.1

To read the logs:

```
supabase functions logs ups-address
```

4.4 Switch to production

```
supabase secrets set UPS_BASE=https://onlinetools.ups.com
```

5 Client setup

Where: the `Settings` tab → `Residential / Commercial Classification`.

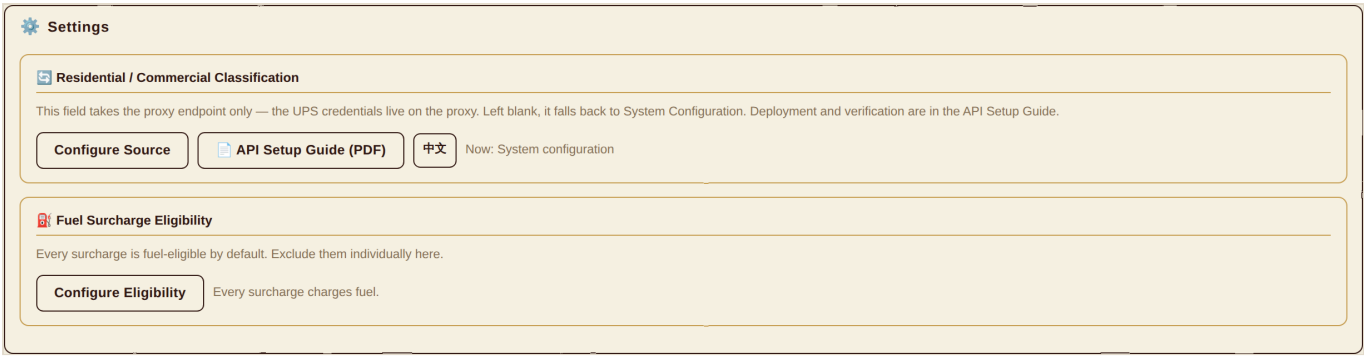


Figure 5-1 The Settings tab. Residential / Commercial Classification is the block this guide concerns; it is configured and saved inside its own panel — there is no shared save button for the page.

> **Where this document comes from**
>
> The **"API Setup Guide (PDF)"** button on the same row gives you the edition matching the interface language; the narrow button beside it (**EN** or 中文) gives the other one, so you never have to switch the whole interface to send a colleague the other edition. English is `UPS-API-Setup-Guide-EN.pdf`, Chinese is `UPS-API-Setup-Guide.pdf`; both must sit in the same folder as `index.html`. If only one was deployed, the button hands over that one and says so; if neither was, it says where the file belongs rather than doing nothing.

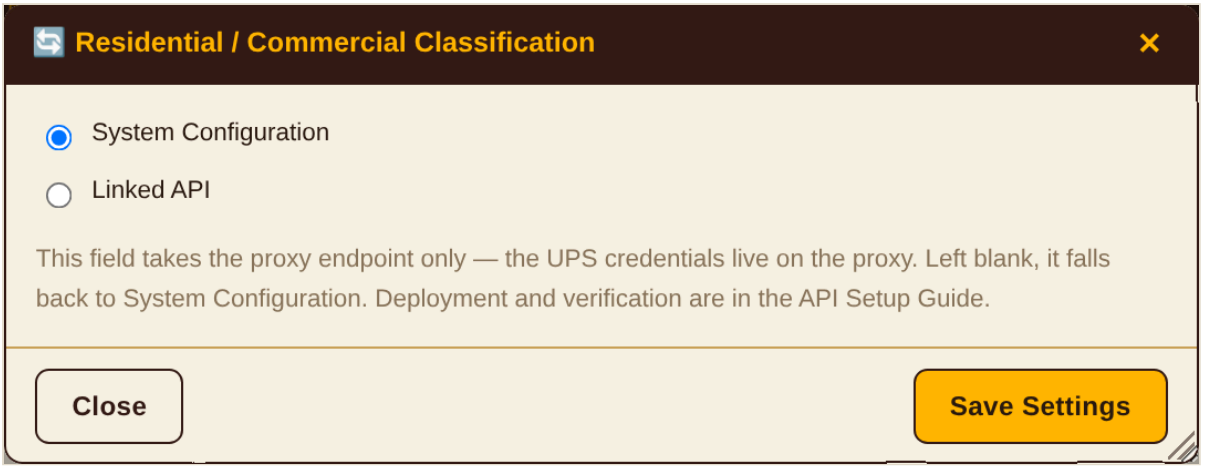


Figure 5-2 The default is System Configuration — billing codes, service name and invoice history.

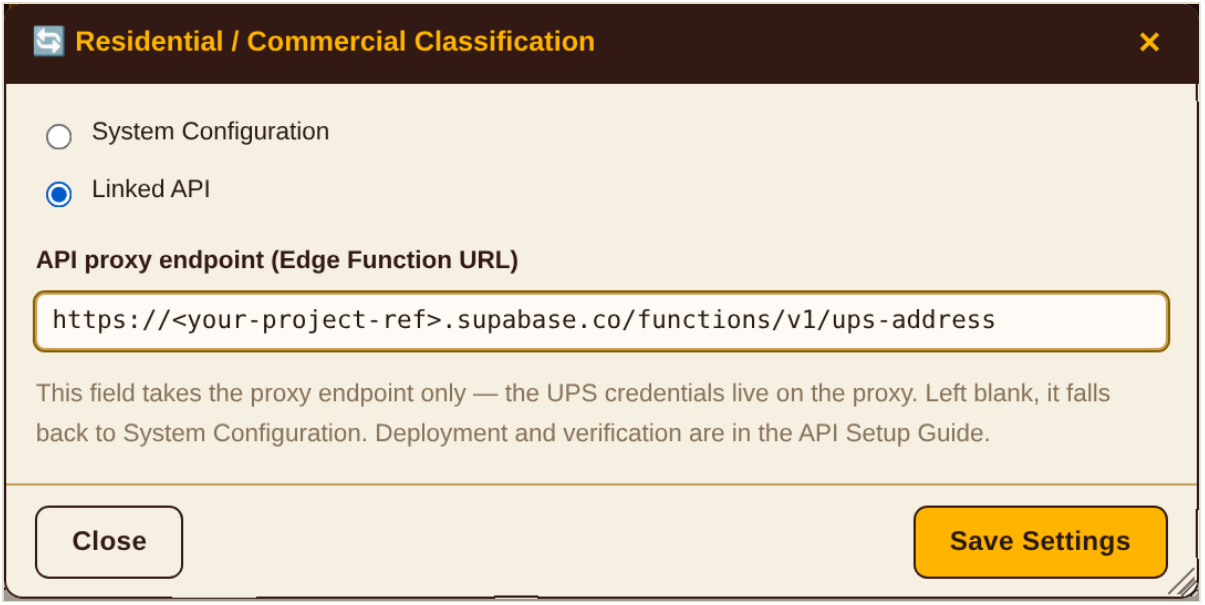


Figure 5-3 Choose Linked API, paste the endpoint you deployed in 3.4, then Save Settings. This field takes the proxy URL only; the UPS credentials live on the proxy (see 1.3).

How the panel behaves

A change inside the panel applies to the in-memory settings the moment you click, and pricing reflects it at once — but nothing is written yet. **Close discards and reverts; Save Settings writes and closes.**

The summary line shows what is actually in force

If Linked API is selected but the endpoint is blank, both the block summary and the radio show System Configuration — because with no endpoint there is nothing to call, and the built-in logic is what actually runs. Showing the setting rather than the behaviour would let someone believe the API was live when it was not.

6 Interface reference

This section defines the proxy endpoint, for anyone integrating against it directly or debugging at the request level. Ordinary setup does not require it.

6.1 Request

Item	Value
Method	`POST`. Anything else returns `405` ; `OPTIONS` serves the CORS preflight
Path	`/functions/v1/ups-address`
Required headers	`Authorization: Bearer <access_token>`
	`Content-Type: application/json`
CORS	Origin `*`, headers `authorization, content-type`

Body fields follow. At least one of `address` or `postal` is required, otherwise the call returns `400` .

Field	Type	Detail
`address`	string	Street address, first line
`city`	string	City
`state`	string	State code
`postal`	string	Postal code; the proxy takes the first five characters
`country`	string	Country code, defaults to `US`
`name`	string	Recipient name, optional; becomes UPS's `ConsigneeName`

6.2 Response

```
{ "class": "R" | "C" | "U", "description": "<UPS classification text>" }
```

class	UPS code	Meaning
`R`	`AddressClassification.Code = 2`	Residential
`C`	`AddressClassification.Code = 1`	Commercial
`U`	`AddressClassification.Code = 0 or absent`	UPS could not tell

Failures return `{"error": "<message>"}` ; status codes are in 8.1. A `502` also carries `detail` , the first 300 characters of the UPS response body.

> **The classification arrives in one of two places**
>
> UPS's `XAVResponse` may put `AddressClassification` at the top level, or under the first `Candidate`. The proxy reads both, in that order, so a difference in placement never turns into a `U` .

6.3 Client behaviour

The client calls this endpoint before each repricing run. It behaves as follows.

Item	Behaviour
Deduplication	Addresses in a batch are normalised and grouped; identical addresses are called once
Concurrency	Six at a time, to stay under the proxy's and UPS's rate limits
Cache	Successful answers are kept, keyed by address; later runs do not call again
Failures	**Failures are never cached.** That run falls back to the built-in logic; once the configuration is fixed, the next run retries automatically
No endpoint set	No request is sent at all. Everything falls back to the built-in logic, and the screen says the endpoint is blank

> **Why failures must not be cached**
>
> If a failed call were recorded as `U`, then a mistyped endpoint, an undeployed function or a missing permission would be written down permanently as "cannot tell" — and fixing the configuration would not undo it, because nothing would retry. Clearing it would mean wiping browser storage by hand. That behaviour is ruled out deliberately.

6.4 The proxy's call to UPS

Item	Value
Endpoint	`{UPS_BASE}/api/addressvalidation/v2/3`
requestoption	3 (Address Validation + Address Classification)
Authorisation	`client_credentials`, via `/security/v1/oauth/token`
Token reuse	Held at the function's module level and reused until 60 seconds before expiry
Trace headers	`transId` (a UUID per request) and `transactionSrc: ups-reprice`

7 Known limits and what to weigh

7.1 Sign-in must use Supabase accounts

> **In local-account mode the endpoint will not work**
>
> The proxy uses `auth.getUser()` to confirm the caller is a real Supabase user. **If your system runs in local-account mode** — accounts written into `index.html`, no Supabase session created — every request returns `401` and pricing falls back to the built-in logic, so the setting in section 5 has no effect however it is filled in.

This is a condition on the authentication layer, not on sections 3 and 4: what those verify is the proxy and the UPS credentials, which are independent of how the client signs in. So do them first even if the authentication model has yet to change — what they produce is reusable infrastructure that a later switch does not invalidate. The section 5 setting takes effect once sign-in uses Supabase accounts.

7.2 Per-account credentials

If each user must draw on their own UPS quota, the proxy has to look the credentials up by caller: keep them in a table only the server can read, and have the proxy fetch the row belonging to the authenticated user. The endpoint URL stays shared. This too depends on there being a real user identity.

7.3 Before you turn it on

> **The API outranks every other source**
>
> Inside `priceShipment()`, an answer from the API takes precedence over billing codes and invoice history. Any shipment the API can classify is classified by the API — including shipments whose invoice already carries a residential code.

So what changes is not a handful of previously ambiguous shipments; it is every shipment the API can classify, and the priced total moves with them.

Suggested approach: turn it on for one account first. Reprice the same invoice period with it on and with it off, compare the totals and the residential / commercial counts, and see how often UPS's classification disagrees with the billing codes. Then decide whether to roll it out.

8 Appendix — troubleshooting

On-screen message	Cause and what to do
"Linked API" is selected with no proxy endpoint	The endpoint field is blank — see section 5
API proxy call failed (HTTP 401)	No valid authenticated session — see 7.1
API proxy call failed (HTTP 404)	Wrong endpoint URL, or the function is not deployed — see 3.4
API proxy call failed (HTTP 500)	UPS credentials not set on the proxy — see 3.3
N addresses failed	Some requests failed; those fall back to the built-in logic. **Failures are not cached, so the next run retries them**
The API returned no classification	UPS answered U (cannot tell) for every address in the batch
"Linked API" is selected but the panel still shows System Configuration	The endpoint is blank; the panel shows what is actually in force — see section 5

8.1 Status codes at a glance

Status	Layer	Go to
`404`	Function not deployed	2.4
`401`	Caller not authenticated	7.1
`500`	UPS credentials not set on the proxy	2.3
`502 UPS 401`	Wrong UPS credentials	2.1
`502 UPS 403`	UPS product not subscribed	2.1
`200`	Normal	—